

Chapter 3: Determinants

MATH 3260 Linear Algebra
Sections 3.1–3.4

Chapter goals

Students will compute determinants using formulas, cofactors, and elementary operations; use determinant properties to analyze products, inverses, and transposes; and apply determinants to inverse formulas, Cramer’s Rule, area, volume, lines, and planes.

Original-example policy

All worked examples and numerical values in these notes are original. Themes involving the Spurs, hiking, travel, and music are used selectively. Any prices or coordinates are fictional teaching models.

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1 The Determinant of a Matrix

Learning objectives

Compute 2×2 determinants; interpret signed area scaling; find minors and cofactors; expand along rows or columns; and evaluate triangular determinants.

1.1 The 2×2 determinant

Definition

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the determinant is

$$\det(A) = ad - bc.$$

Original example: Spurs-inspired values

For

$$A = \begin{bmatrix} 9 & 1 \\ 20 & 21 \end{bmatrix},$$

we obtain

$$\det(A) = 9(21) - 1(20) = 169.$$

For

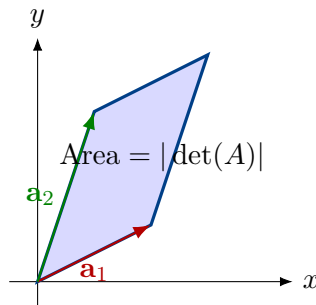
$$B = \begin{bmatrix} 4 & 6 \\ 2 & 3 \end{bmatrix},$$

we obtain $\det(B) = 0$, while

$$\det \begin{bmatrix} -2 & 5 \\ 7 & 1 \end{bmatrix} = -37.$$

1.2 Geometric meaning

The columns of a 2×2 matrix form a parallelogram. Its area is $|\det(A)|$. The sign records whether orientation is preserved or reversed.



1.3 Minors and cofactors

Minor and cofactor

The minor M_{ij} is obtained by deleting row i and column j and taking the determinant of the remaining matrix. The cofactor is

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

The sign pattern begins

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}.$$

All minors and cofactors

Let

$$H = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 3 & 2 \\ 4 & 0 & 1 \end{bmatrix}.$$

For example,

$$M_{12} = \det \begin{bmatrix} -1 & 2 \\ 4 & 1 \end{bmatrix} = -9, \quad C_{12} = 9.$$

The minor and cofactor matrices are

$$[M_{ij}] = \begin{bmatrix} 3 & -9 & -12 \\ 1 & 2 & -4 \\ 2 & 4 & 7 \end{bmatrix}, \quad [C_{ij}] = \begin{bmatrix} 3 & 9 & -12 \\ -1 & 2 & 4 \\ 2 & -4 & 7 \end{bmatrix}.$$

1.4 Cofactor expansion

Expansion along row i or column j gives

$$\det(A) = \sum_{k=1}^n a_{ik} C_{ik} \quad \text{or} \quad \det(A) = \sum_{k=1}^n a_{kj} C_{kj}.$$

Choose a row or column containing many zeros.

For H , expansion along row 1 gives

$$\det(H) = 2C_{11} + 1C_{12} + 0C_{13} = 2(3) + 1(9) = 15.$$

Sparse 4×4 expansion

Let

$$Q = \begin{bmatrix} 2 & 1 & 0 & 3 \\ 0 & -1 & 0 & 4 \\ 5 & 2 & 7 & 1 \\ 1 & 0 & 0 & 2 \end{bmatrix}.$$

The third column has only one nonzero entry, so

$$\det(Q) = 7C_{33}.$$

The corresponding minor is

$$\begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 1 & 0 & 2 \end{bmatrix}.$$

Expanding it along row 2 gives

$$(-1)(4 - 3) + 4 \left[-\det \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix} \right] = -1 + 4 = 3.$$

Therefore,

$$\det(Q) = 7(3) = 21.$$

1.5 Triangular matrices

Triangular determinant theorem

The determinant of an upper or lower triangular matrix is the product of its diagonal entries.

For

$$T = \begin{bmatrix} 3 & 0 & 0 & 0 \\ -2 & 5 & 0 & 0 \\ 1 & 4 & -2 & 0 \\ 7 & 1 & 6 & 4 \end{bmatrix},$$

we have

$$\det(T) = 3(5)(-2)(4) = -120.$$

Common mistakes

Use $ad - bc$ only for 2×2 matrices. During cofactor expansion, include the alternating signs and multiply each entry by its cofactor.

Closing note

A determinant can be computed in several equivalent ways. Efficient work depends on recognizing zeros, triangular structure, and the cofactor sign pattern.

1.6 Definitions and terminology

Term	Meaning
Determinant	Scalar associated with a square matrix.
Signed area scale	Geometric meaning of a 2×2 determinant.
Minor M_{ij}	Determinant after deleting row i and column j .
Cofactor C_{ij}	$(-1)^{i+j} M_{ij}$.
Cofactor expansion	Sum of entries times cofactors along a row or column.
Upper/lower triangular	Zeros below/above the main diagonal.
Diagonal matrix	Zeros above and below the main diagonal.

2 Determinants and Elementary Operations

Learning objectives

Track determinant changes under row and column operations, reduce matrices to triangular form, and recognize structures that force a zero determinant.

Effects of elementary operations

For row or column operations:

1. Interchanging two rows changes the sign of the determinant.
2. Adding a multiple of one row to another leaves the determinant unchanged.
3. Multiplying one row by c multiplies the determinant by c .

2.1 Detailed row-reduction calculation

Let

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 4 & 3 & 0 \\ -2 & 2 & 5 \end{bmatrix}.$$

Use only row replacements:

$$R_2 \leftarrow R_2 - 2R_1, \quad R_3 \leftarrow R_3 + R_1, \quad R_3 \leftarrow R_3 - 3R_2.$$

These operations do not change the determinant:

$$\begin{bmatrix} 2 & 1 & -1 \\ 4 & 3 & 0 \\ -2 & 2 & 5 \end{bmatrix} \sim \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 2 \\ -2 & 2 & 5 \end{bmatrix} \sim \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 2 \\ 0 & 3 & 4 \end{bmatrix} \sim \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & -2 \end{bmatrix}.$$

Therefore,

$$\det(A) = 2(1)(-2) = -4.$$

2.2 Row swaps and row scaling

If one row is swapped, the determinant changes sign. If one row is multiplied by 5, the determinant is multiplied by 5. Keep a written accounting record when several operations are used.

2.3 Column operations and zero determinants

For

$$D = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 8 & 5 \\ 2 & 4 & 7 \end{bmatrix},$$

the second column is twice the first. The operation

$$C_2 \leftarrow C_2 - 2C_1$$

creates a zero column, so $\det(D) = 0$.

Conditions that yield zero

The determinant is zero if a row or column is zero, if two rows or columns are equal, or if one row or column is a multiple of another. More generally, linear dependence among rows or columns forces determinant zero.

Redundant data features

If one recorded feature is exactly determined by other features, the feature columns are dependent. The determinant is zero, and the feature matrix cannot be inverted.

Closing note

Row reduction can evaluate a determinant efficiently, but every swap and scaling operation must be recorded. Row replacement is especially useful because it leaves the determinant unchanged.

2.4 Definitions and terminology

Term	Meaning
Row interchange	Swap two rows; determinant changes sign.
Row replacement	Add a row multiple to another row; determinant unchanged.
Row scaling	Multiply one row by c ; determinant multiplied by c .
Column-equivalent	Related by elementary column operations.
Zero determinant	Determinant equal to zero; matrix is singular.
Redundant information	A row or column depends on others.

3 Properties of Determinants

Learning objectives

Use determinant rules for products, scalar multiples, inverses, transposes, and powers; connect nonzero determinant with invertibility.

3.1 Products

Product theorem

For square matrices of the same order,

$$\det(AB) = \det(A) \det(B).$$

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 \\ 4 & 2 \end{bmatrix}.$$

Then $\det(A) = 5$, $\det(B) = 6$, and

$$\det(AB) = 30.$$

Indeed,

$$AB = \begin{bmatrix} 6 & 0 \\ 13 & 5 \end{bmatrix}$$

has determinant 30.

3.2 Scalar multiples

For an $n \times n$ matrix,

$$\det(cA) = c^n \det(A).$$

If

$$C = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix},$$

then $\det(C) = 6$ and

$$\det(2C) = 2^3(6) = 48.$$

3.3 Inverses and nonsingularity

A square matrix is invertible exactly when its determinant is nonzero. If A is invertible,

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

For the matrix above, $\det(A^{-1}) = 1/5$.

Equivalent nonsingularity conditions

For an $n \times n$ matrix A , the following are equivalent: A is invertible; $A\mathbf{x} = \mathbf{b}$ has one solution for every $\mathbf{b}$; $A\mathbf{x} = \mathbf{0}$ has only the trivial solution; A is row-equivalent to I_n ; A is a product of elementary matrices; and $\det(A) \neq 0$.

3.4 Transpose and powers

For every square matrix,

$$\det(A^T) = \det(A).$$

Repeated multiplication gives

$$\det(A^k) = \det(A)^k.$$

For $\det(A) = 5$, we have $\det(A^4) = 625$.

Composition of transformations

If one transformation scales signed area by 6 and a second scales it by 5, their composition scales signed area by 30. This is the geometric meaning of $\det(AB) = \det(A)\det(B)$.

Closing note

Determinant properties often avoid expensive matrix calculations. Compute the determinants of the factors instead of multiplying large matrices first.

3.5 Definitions and terminology

Term	Meaning
Product rule	$\det(AB) = \det(A)\det(B)$.
Scalar-multiple rule	$\det(cA) = c^n \det(A)$.
Nonsingular	Invertible; determinant nonzero.
Singular	Not invertible; determinant zero.
Inverse determinant	$\det(A^{-1}) = 1/\det(A)$.
Transpose determinant	$\det(A^T) = \det(A)$.
Power rule	$\det(A^k) = \det(A)^k$.

4 Applications of Determinants

Learning objectives

Use the adjugate to find inverses, solve systems using Cramer's Rule, and apply determinants to area, volume, collinearity, coplanarity, lines, and planes.

4.1 The adjugate and inverse

The adjugate is the transpose of the cofactor matrix:

$$\text{adj}(A) = [C_{ij}]^T.$$

If $\det(A) \neq 0$, then

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A).$$

For

$$M = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix},$$

the cofactor matrix and adjugate are

$$C = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix}, \quad \text{adj}(M) = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix}.$$

Because $\det(M) = 2$,

$$M^{-1} = \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix}.$$

4.2 Cramer's Rule

For $A\mathbf{x} = \mathbf{b}$ with $\det(A) \neq 0$, replace column i by $\mathbf{b}$ to form A_i . Then

$$x_i = \frac{\det(A_i)}{\det(A)}.$$

Concert merchandise prices

At a fictional Backstreet Boys fan event,

$$\begin{cases} p + w = 29, \\ w + t = 41, \\ p + t = 30. \end{cases}$$

The coefficient determinant is $D = 2$, while

$$D_p = 18, \quad D_w = 40, \quad D_t = 42.$$

Thus

$$p = 9, \quad w = 20, \quad t = 21.$$

4.3 Area and collinearity

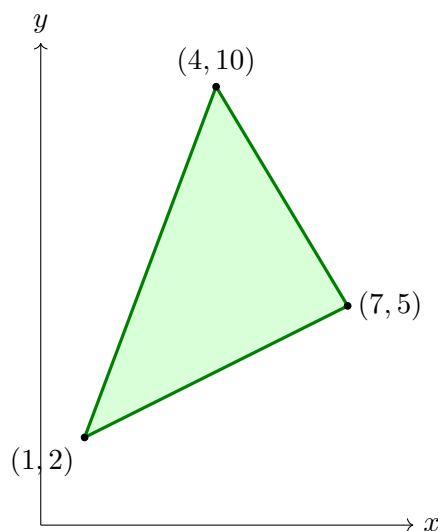
The triangle area formula is

$$\text{Area} = \frac{1}{2} \left| \det \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix} \right|.$$

For hiking landmarks $(1, 2)$, $(7, 5)$, and $(4, 10)$,

$$\text{Area} = \frac{1}{2} |39| = 19.5.$$

The points are collinear exactly when the determinant is zero.



4.4 Equation of a line

The line through (x_1, y_1) and (x_2, y_2) satisfies

$$\det \begin{bmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{bmatrix} = 0.$$

For $(2, 5)$ and $(8, 17)$, expansion gives

$$y = 2x + 1.$$

4.5 Volume and coplanarity

For four points in space,

$$V = \frac{1}{6} \left| \det \begin{bmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{bmatrix} \right|.$$

For $(0, 0, 0)$, $(2, 0, 0)$, $(0, 3, 0)$, and $(0, 0, 5)$,

$$V = 5.$$

Four points are coplanar exactly when the determinant is zero.

4.6 Equation of a plane

For points

$$P_1 = (1, 0, 2), \quad P_2 = (3, 1, 0), \quad P_3 = (0, 2, 1),$$

we have

$$(P_2 - P_1) \times (P_3 - P_1) = (3, 4, 5).$$

Therefore, the plane equation is

$$3x + 4y + 5z = 13.$$

Equivalently, the determinant

$$\det \begin{bmatrix} x & y & z & 1 \\ 1 & 0 & 2 & 1 \\ 3 & 1 & 0 & 1 \\ 0 & 2 & 1 & 1 \end{bmatrix} = 0$$

defines the same plane.

Closing note

Determinants connect algebraic structure with geometry: they test invertibility, solve square systems, and encode area, volume, lines, and planes in compact formulas.

4.7 Definitions and terminology

Term	Meaning
Cofactor matrix	Matrix $[C_{ij}]$.
Adjugate	Transpose of the cofactor matrix.
Adjugate inverse formula	$A^{-1} = \text{adj}(A) / \det(A)$.
Cramer's Rule	$x_i = \det(A_i) / \det(A)$.
Collinear	Points on one line.
Coplanar	Points in one plane.
Determinant line equation	3×3 determinant equal to zero.
Determinant plane equation	4×4 determinant equal to zero.
Tetrahedron volume	One sixth of an absolute 4×4 determinant.

Chapter summary

Connected storyline

cofactors $\longrightarrow$ row-operation evaluation
 $\longrightarrow$ invertibility and geometric applications.

1. Use $ad - bc$ only for 2×2 matrices.
2. Choose sparse rows or columns for cofactor expansion.
3. Track every swap and scaling during row reduction.
4. Remember $\det(cA) = c^n \det(A)$.
5. Connect $\det(A) = 0$ with singularity.
6. Use absolute values for area and volume.